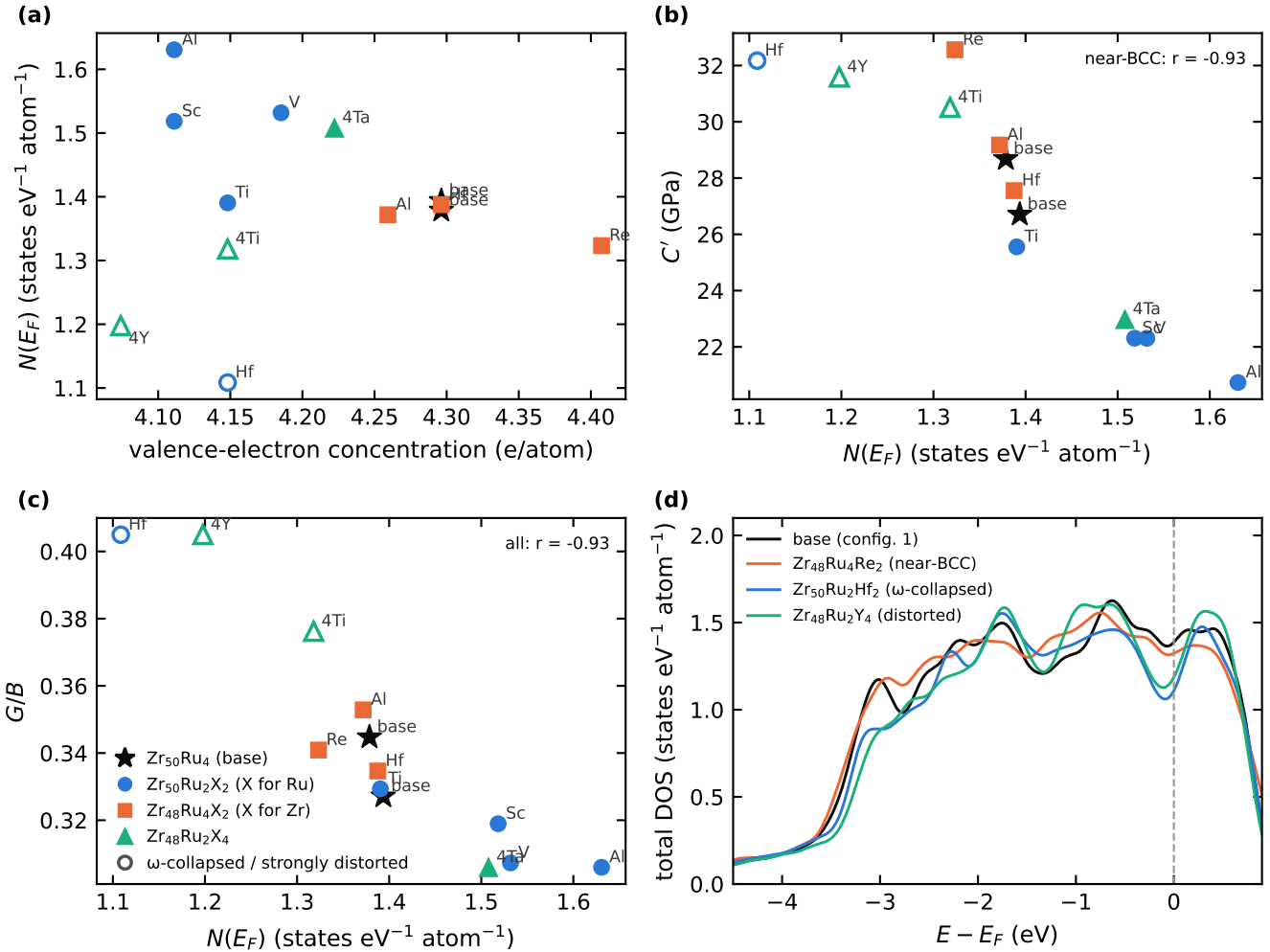
(a)
$N(E_F)$ (states eV^{-1} atom^{-1})
valence-electron concentration (e/atom)
Al
Sc
V
4Ta
Ti
Al
base
base
4Ti
Re
4Y
Hf
(b)
near-BCC: r = -0.93
C' (GPa)
$N(E_F)$ (states eV^{-1} atom^{-1})
Hf
Re
4Y
4Ti
Al
base
Hf
base
Ti
4Ta
V
Al
(c)
all: r = -0.93
G/B
$N(E_F)$ (states eV^{-1} atom^{-1})
Zr$_{50}$Ru$_4$ (base)
Zr$_{50}$Ru$_2$X$_2$ (X for Ru)
Zr$_{48}$Ru$_4$X$_2$ (X for Zr)
Zr$_{48}$Ru$_2$X$_4$
ω-collapsed / strongly distorted
(d)
total DOS (states eV^{-1} atom^{-1})
$E - E_F$ (eV)
base (config. 1)
Zr$_{48}$Ru$_4$Re$_2$ (near-BCC)
Zr$_{50}$Ru$_2$Hf$_2$ (ω-collapsed)
Zr$_{48}$Ru$_2$Y$_4$ (distorted)